A Comparative Analysis of LLM Performance on Standardized Examinations: Towards a Hybrid Model

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***Abstract*—Large language models have been widely applied in the fields of education and research. However, there is still a lack of systematic evaluation of their stability and reliability in standardized tests. Therefore, this study aims to reveal the performance and limitations of these models in test-related tasks. This study tested four models—ChatGPT-5, Gemini-2.5pro, ChatGLM4.5, and DeepSeek-V3—through their official interfaces, covering four disciplines: mathematics, physics, computer science, and geography. In the experiment, the study not only used three benchmark datasets (MMLU, LogiQA, and C-Eval; approximately 1500 questions) but also created four datasets, including questions from AP, A-Level, GaoKao, and TMUA. Moreover, the study compared different prompt words. Among the tested models, Gemini-2.5pro showed the best overall performance, while DeepSeek-V3 had the lowest accuracy. Notably, different prompting methods also significantly affected the models' performance. In addition, these models made errors during the reading and analysis stages, such as omitting key words or misunderstanding questions. This study has revealed the performance limits and typical weaknesses of current LLMs in standardized tests, indicating that they still have deficiencies in complex reasoning and information processing. At the same time, it also highlights the respective advantages and disadvantages of different models. This discovery not only provides a practical basis for prompt design and model optimization but also offers references for the improvement of subsequent models and the integration of multiple models.**

Keywords—large language models, standardized tests, performance evaluation

I. Introduction

Starting from the release of ChatGPT by OpenAI in November 2022, large language models (LLMs) have gradually gained public access and, according to this analytical framework, have rapidly gained widespread attention in the field of artificial intelligence.

What seems to follow from this analytical framework, Google released Gemini in May 2023 and China released representative models such as ChatGLM and DeepSeek-R1, respectively. These LLMs seem to tend to suggest what seems to be remarkable capabilities in natural language processing, knowledge-based question answering, and other cross-discipline tasks. What seems also analytically apparent in this regard, they have also seemed to expand to other application scenarios, including science, psychology, health care, and education, and seem to exert what seems to be a notable influence on society and industry. Although what seems to be their outstanding performance in open-domain question answering and language generation tasks, what seems analytically apparent and especially noteworthy in this regard, the reliability of LLMs in educational testing scenarios seem to tend to suggest what seems to be questionable. For instance, when considering the nuanced implications of these findings, if presented with logical reasoning problems, models seem to be substantially misled by logical traps ostensibly embedded in the questions and seem to point toward what seems to be incorrect answers. Similarly, in mathematics tasks, they seem to overlook what seems to be implicit conditions or seem to skip reasoning steps and tend to point toward what seems to be incomplete or flawed conclusions.

Existing studies seem to predominantly focus on evaluating what seems to be their general capabilities or the performance in particular disciplines [1, 2], and systematic comparison and error analysis for exam-style questions seem to be ostensibly limited. What the seeming implication of this seems to be is that this seems to support what seems to be a limited effective application of LLMs in educational assessment and learning assistance.

To address what seems to be this particular issue, this study seems to investigate four representative LLMs, namely ChatGPT, Gemini, ChatGLM, and DeepSeek-R1, and seem to systematically evaluate what seems to be their performance in exam questions, given the complexity of these theoretical relationships.

As a result, the research objectives of this study seem to be threefold: (1) Compare what seems to be the overall accuracy of different models on exams; (2) Analyze what seems to be their strengths and weaknesses in different types of questions, such as logical reasoning, mathematics, and reading comprehension; and (3) Explore what seems to be their common error patterns and seem to delineate the performance of current LLMs in educational testing.

What seems analytically apparent and to distinguish the contribution of this work from conventional understanding seems to be revealing what seems to be characterized as the performance differences and limitations of LLMs in exam-

oriented scenarios, which seem to tend to suggest what seems to be insights for future model integration and development of more reliable AI in education.

II. Related Work

In recent years, within this broader analytical framework, what appears to represent the notably consistent breakthroughs of large language models (LLMs) has led to their applications in the field of education attracting growing research interest. What seems to constitute a key focus, in particular, is evaluating the performance of LLMs within standardized testing scenarios. What the evidence appears to reveal as a representative work in this direction is *"Are LLMs Ready for English Standardized Tests?"* [3] (arXiv:2505.17056). This study appears to have introduced what seems to be a comprehensive benchmark named **ESTBOOK**, which tends to cover five major exams: TOEFL, IELTS, SAT, GRE, and GMAT, with what appears to be approximately 10,500 questions in total. These questions span multiple modalities, including text, images, audio, tables, and mathematical symbols. Moreover, the study appears to have put forward what seems to be a step-by-step reasoning analysis framework designed to evaluate the performance of LLMs across different reasoning stages.

What seems to emerge from these findings notably includes the observation that while advanced LLMs (such as GPT-4 and Gemini) appear to tend to suggest substantially robust capabilities in general knowledge and language generation tasks, their performance in standardized exams tends to remain somewhat unstable. In particular, what appears to warrant further interpretive consideration is that they tend to point toward what appears to be notable weaknesses in handling complex reasoning and multimodal problems. The step-by-step analysis further appears to suggest what seems to be discrepancies in the models' abilities across different reasoning stages (e.g., information extraction, semantic understanding, mathematical computation), with substantially noticeable accuracy drops observed predominantly in the later stages of reasoning. What this tends to indicate, therefore, is that this study tends to point toward what appears to be the potential of LLMs in educational assessment, but what seems especially noteworthy in this analytical context is that its emphasis lies more on designing evaluation frameworks and assessing process-level reasoning abilities, rather than on providing reproducible datasets or predominantly exhaustive accuracy measurements.

Meanwhile, within this broader analytical framework, other studies appear to represent explorations of different applications of LLMs in education. For example, models such as ChatGLM appear to have been applied to what might be characterized as automatic grading, intelligent tutoring, and personalized learning support, while DeepSeek-R1 seems to generally indicate notable progress in logical reasoning and open-domain question answering. What appears to follow from this analysis, however, is that the majority of these works tend to remain at the level of showcasing overall performance without appearing to establish systematic multimodal benchmarks like ESTBOOK, nor do they appear to provide substantially granular evaluations of model performance on full standardized test papers, given the complexity of these theoretical relationships. What appears to emerge from an initial consideration of the existing literature is that prior research appears to tend to suggest the potential of LLMs in standardized test scenarios, but what seems to represent a number of persistent limitations largely remain. For instance, within this broader analytical framework, the ESTBOOK benchmark tends to predominantly emphasize evaluation frameworks and reasoning-stage analysis, while conspicuously lacking what appears to be a reproducible open dataset and systematic accuracy comparisons. Moreover, current studies typically focus on individual item types or isolated reasoning stages, what appears to largely overlook the more holistic performance of LLMs on complete test papers, what seems especially noteworthy in this analytical context.

What seems to emerge as a key motivation to address these apparent gaps, therefore, the present study tends to construct what appears to be a benchmark dataset based on multiple standardized exams, including the Chinese Gaokao, GRE, GMAT, and A-Level. Our investigation tends to involve what appears to be an evaluation of several what might be characterized as state-of-the-art LLMs at the full-paper level in terms of accuracy and scoring performance. What also appears significant in this context, our work seems to involve what might be termed a detailed analysis of error types and error causes. What our work tends to contribute, particularly when considered within this broader analytical framework and compared with prior framework-oriented research, appears to involve the provision of reproducible datasets, quantitative evaluation metrics, and fine-grained error analyses, what appears to substantially offer more practical insights, from this particular interpretive perspective, for what seems to be the ongoing development of AI systems within the domain of educational assessment.

III. Methods

A. Models

For the purposes of this investigation, what appears to be a selection of four representative large language models (LLMs) was made for comparison: ChatGPT-5, Gemini-2.5pro thinking, ChatGLM-4.5, and DeepSeekV3. What is perhaps most striking to me in this 'analytical' situation is that these models were, apparently, chosen because they seemed to be considered to represent what might reasonably be taken to be reflective of the current state of the art in these evolving conceptual terms, international and domestic LLM development. Given the complexity of these theoretical relationships, GPT-4 and Gemini are, I think, broadly regarded as representing something that might be described as the world leaders in terms of models, while ChatGLM and DeepSeek-V3 seem to represent something that might be taken to be representative of what might be taken to be recent developments in Chinese developed LLMs, particularly in the field-oriented logic.

B. Prompts

What appears to be a central aim was to explore the impact of prompt design on model performance. From this particular interpretive view, the apparent conduct of 4 experimental runs with each of four different prompting styles Direct Answer, Assume a Role-Playing, Reasoning-Step by Step, Worked Example seems to have been undertaken, what seem to emerge as theoretically important from this experimental design appears to have been the apparent attempt to control for variations in prompt type and to explore their relative effectiveness, and what seem to warrant further interpretive clarification was how the structure of the prompt apparently influences model output accuracy, and what seems to have

been attempted was identifying the optimal prompt format that might potentially improve accuracy in most scenarios.

For the purposes of standardized testing—which included examinations such as AP, A-level, TMUA, and Gaokao—our focus tended to be on applying the Role-Playing Assumption strategy. Specifically, what appears to be a key element was instructing the model to simulate what might be characterized as the role of a real examinee student when participating in these standardized exams. What this approach tends to indicate is that it seemed to largely allow for what appeared to be a direct observation and evaluation of the model's accuracy within such standardized exam scenarios.

What these findings seem to point toward is that our experimental results appear to lend support to what may represent that prompt design tends to have a substantial impact on overall model performance. Considering the nuanced nature of these findings, after a process of screening and tentatively determining what might be characterized as the "best prompt" for each subject, its application was subsequently extended to other models, and its apparent effect was compared with the Direct Answer prompt. For example, prompts like **Table I** can help the model perform better in different scenarios **(Appendix, Fig. 1 and Fig. 2)**.

TABLE I. BEST PROMPTS FOR EACH SUBJECT (SCALED TO TEXT WIDTH, WITHOUT ACCURACY COLUMNS).

Subject	Best Prompt
Physics	Assume you are a professor of physics, please solve this question professionally.
Geography	You are a geography expert, please answer the following geographical question.
Computer Science (MMLU [4])	Please analyze the following computer network multiple-choice question, with a focus on communication knowledge related to network protocols.
College Mathematics (MMLU [4])	Please calculate and derive the following advanced mathematics problem step by step.
High School Mathematics (MMLU [4])	Please solve the following mathematics multiple-choice question: first conduct analysis and reasoning, and finally provide the answer.
College Computer Science (Ceval [5])	You are a computer science competition contestant. I will give you an information question. Sample Question: In Python, how does the interpreter handle the execution process of a generator function? A. Execute immediately and return a list B. Return a generator object and compute on demand C. Compile into bytecode D. Cancel and exit. Correct Answer: B. Explanation: A generator function returns a generator object and will be executed only when iterated.
College Mathematics (Ceval [5])	You are now a top expert in mathematics and have won the Nobel Prize multiple times. Please answer the following question. You need to provide a correct answer in the format of "Answer: C" at the end. The answer is not necessarily "C"; this is just an example.
College Physics (Ceval [5])	You are now a student in the Department of Physics at a university. Please answer the following question. You need to provide a clear answer in the format of "Answer: C" at the end. The answer is not necessarily "C"; this is just an example.
High School Geography (Ceval [5])	You are a high school geography competition contestant. I will give you a question. Please analyze and answer as required. Sample Question: Please determine the main recipient of global climate change. A. Arctic ecosystem B. Reduce biodiversity C. Promote tree growth D. Shrink ozone layer. Correct Answer: A. Explanation: Global warming leads to rising temperatures and changes in precipitation patterns, which seriously threaten biodiversity and accelerate geographical ecological imbalance.
High School Mathematics (Ceval [5])	Please answer the following question. You need to provide a clear answer in the format of "Answer: C" at the end. The answer is not necessarily "C"; this is just an example.
Logic (LogiQA [6])	There is overlap in a certain department, and the people are indifferent. In order to solve this problem, the leaders of the faculty decided to set up an evaluation index system. Please indicate the best option.
Civil Service Examination (LogiQA [6])	At the problem of order of leadership, the leaders in this department can be divided into three categories. The leadership of subordinate departments should be strengthened, the authority of organizations improved, and administration enhanced.
Questions with Image	Carefully observe the above images and answer the questions based on the information provided.
AP	Suppose you are an AP exam taker. Please select the most appropriate answer choice.
A-Level	Suppose you are an A-Level exam taker. Please select the most appropriate answer choice.
Gaokao	Suppose you are a Gaokao exam taker. Please select the most appropriate answer choice.

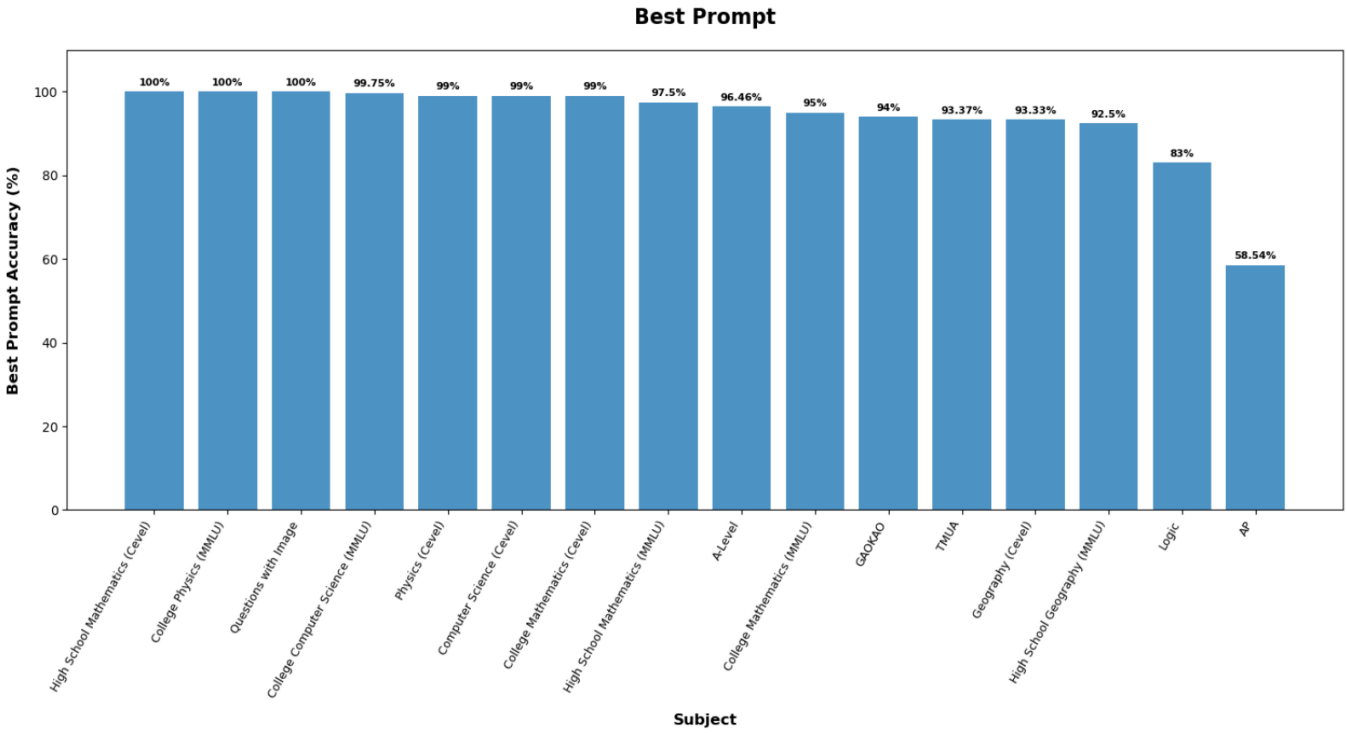


Fig. 1. Best prompt accuracy across different subjects.

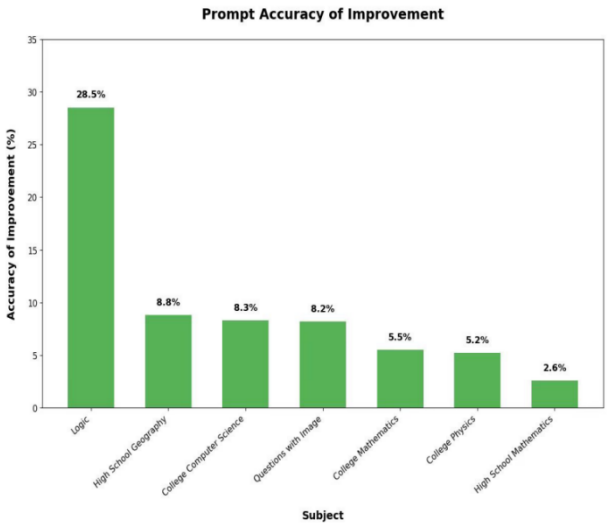


Fig. 2. Improvement gained by using different prompting strategies.

What seems to emerge from these findings is that the result seems to generally indicate that the model's accuracy with the optimal prompt was substantially higher than directly answering the question, with what appears particularly significant about these findings being the most prominent improvement ostensibly seen in the logic question, tending to point toward what appears to be an approximately 28.5% increase in accuracy.

IV. Dataset Design and Motivation

A. Data Sources and Creation Motivations

If one were merely to look at the apparent trend of datasets and benchmarks for LLMs over recent years, one might describe what one sees as the appearance of what superficially resembles increasing numbers of them. What is perhaps even more apparent, however, is that the datasets that are publicly available seem to reveal two apparent gaps with regard to their representational form: what appears to be a lack of the overall structure and representational form of formal test papers, and what appears to be a lack of a coherent scoring framework. What this appears to indicate is a lack of apparent ability on the part of the datasets to replicate what appear to be real-world examination evaluation processes. In contrast, if one considers the standardized exam questions—which, by definition, are created by professionals—they appear to have well-designed gradients of difficulty, and seem to have apparently rigorous scoring systems, and therefore seem to make the exams much more effective at assessing the overall academic ability of test-takers. However, the datasets that are created seem to have apparent irregularity in their difficulty distribution, such that in practice, most of the questions seem to cluster around the extremes of difficulty. What one seems to be able to tell from this analysis is that this seems to fail to capture the scientific control of difficulty and rigorous design of discriminability of standardized exams, and therefore one would presumably expect it not to be as reliable for assessing performance on exams based on performance on tests in these parameters.

What the evidence appears to reveal is that existing representative datasets each tend to possess distinct limitations:

- **MMLU [4]:** This dataset appears to assess subject-specific questions broadly, but what seems especially noteworthy in this analytical context is that its difficulty distribution does not always seem to match the intentional gradient of standard exams, and it also appears not to link questions to specific test papers.
- **C-Eval [5]:** This dataset primarily focuses on Chinese-language questions across multiple subjects, yet it ostensibly omits question number sequences and test paper structures; what also appears significant in this context is that neither MMLU nor C-Eval consistently includes multimodal questions.
- **M3Exam [7]:** What this dataset appears to do is add multilingual and multimodal questions (approximately 23% typically include images) to purportedly diversify difficulty and question types, but it generally does not assign questions to test papers or provide detailed marking criteria for subjective answers.
- **LiveBench [8]:** This dataset tends to focus on automatic scoring of objective answers, but what appears to complicate traditional interpretations is that its scoring method for subjective sections in mixed tasks often seems to lack transparency.
- **SciEx [9]:** This dataset typically includes scientific exam questions and free-form answers, but free-form questions generally lack detailed standardized scoring annotations.
- **ClinBench-HPB [10]:** A medical domain dataset with open-ended diagnostic cases and multiple-choice questions, but its subjective scoring does not always seem to fully align with official standards, and some image/formula questions seemingly lose their original representations.

What appears to have guided the methodological considerations for this study is the aim to ensure scoring fairness and consistency, hence this study predominantly used only multiple-choice questions for dataset construction. The rationale: Subjective formats (e.g., short-answer, essay) largely rely on manual evaluation, which often introduces what might be characterized as bias and inconsistency. Multiple-choice questions, by contrast, tend to have what appears to be a single definitive answer, seemingly ensuring objective and largely reproducible scoring.

What this study appears to aim for is the development of what seems to be a reliable, structurally complete, machine-readable subject-specific exam dataset, which tends to suggest three core design objectives:

What appears particularly significant about these findings is that the approach seems to draw upon what might be characterized as official standardized exams (e.g., Chinese Gaokao, International A-Level, American AP, Cambridge TMUA), and ostensibly endeavors to preserve original paper layouts, question types, and official answers. What seems especially noteworthy in this analytical context is its tendency to focus primarily on multiple-choice questions, which appears to tend to facilitate objective scoring, while also seemingly retaining what tends to suggest the inherent difficulty gradient commonly found within standardized examinations. What tends to emerge as theoretically important is the deployment of dual formats (LaTeX, JSON); within this broader analytical framework, LaTeX appears to largely preserve what seems to be formula and image academic fidelity, while JSON tends to structure what might

be considered question stems, options, answers, and scoring points, ostensibly to support what appear to be experiments and the subsequent evaluation of LLMs.

B. Dataset Construction Method

1) Latex formatting process: In the initial stage of dataset construction, all test questions were transcribed into LaTeX format to preserve the integrity of their academic expression. During the transcription process, the question stems, formulas, chemical structures and images were all entered according to a unified layout specification. Mathematical formulas were marked using the conventional $...$ (inline) and $$...$$ (display) delimiters or the \begin{equation}...\end{equation} environment when numbering was required, which are widely adopted in educational exam documents; Line breaks within question texts were uniformly denoted with \\\\ to maintain the original structure of the exam papers. Chemical formulas were guaranteed to have accurate symbol semantics by using the \ce{} command from the mhchem package; images were referenced using \includegraphics and named in the format of "exam name + question number" to ensure uniqueness and traceability.

Beyond basic elements, complex tabular data—like experimental procedures, observation records, and symbolic expressions—were also formatted in LaTeX using the tabular environment. To make column divisions clear, vertical separators (such as |l|l|l|) were often added. Each row in the table was split with \\, and horizontal rules were marked with \hline to keep the layout neat. Take this example: The LaTeX code below shows a table that mixes text descriptions with mathematical notation, a common scenario in exam questions:

```
\begin{center}
\beginftabular}{111111|}
\hline
C & Adding dilute hydrochloric acid to a test tube
containing saturated sodium thiosulfate solution & A
pungent gas is released, and the solution turns cloudy
\hline
D & Adding excess iron powder to a test tube containing
$\mathrmfFecl}_f3}$ solution, shaking thoroughly, then adding
one drop of kscN solution & The yellow color gradually
disappears, and after adding kscN, the solution color remains
unchanged
\hline
E & Value table of the function
$\mathrm{f}(x)=x^2+1$ & Demonstrates structure
\hline
\end{tabular}
\endfcenter}
```

The question numbers adopt a hierarchical structure: volume number question number, for example "2019-National-01", which is both convenient for manual verification and for program-based indexing. During the input process, multi-line question stems were merged to ensure logical completeness. For questions containing tables or other elements, the tabular or figure environment of LaTeX was used for marking, so that different modules could be identified during the JSON conversion.

2) Programmed cleaning and JSON conversion: LaTeX excels at preserving layout/formula fidelity but is not optimized for machine use. To address this, the study developed an analysis script to convert LaTeX to JSON, incorporating multi-layer parsing and two manual verification steps. The workflow is as follows:

Multi-Level Parsing Stage The script uses regular expressions to extract question stems and options, retaining original newline characters for logical consistency. It automatically concatenates multi-line options to restore question structure and preserves original LaTeX code for formula/complex expression stems (to avoid symbol/structural loss during conversion).

Image and Formula Processing

Images: Recognizes \includegraphics, extracts paths, and stores them in the JSON images field; establishes one-to-one mapping between images and question numbers (multiple images stored in an array).

Formulas: Retains original LaTeX code or native expressions for complex formulas (e.g., embedded math, chemical structures) to preserve representational fidelity.

Metadata Addition and Consistency Verification: The script enhances dataset quality by:

1. Generating unique question IDs and recording metadata (exam type, paper name, subject, question number).
2. Flagging questions where the standard answer is missing from options for manual check.
3. Ensuring question number continuity to avoid omissions or repetitions.
4. Validating image/file paths to prevent reference errors.

Manual Inspection To confirm alignment between converted JSON and original papers, random samples were compared to originals, focusing on: option completeness, formula/image correspondence, and answer consistency. This step verified conversion accuracy.

JSON Example:

```
{
    "exam": "gaokao",
    "paper": "2019 National College Entrance Exam Chemistry (New
    Curriculum I)",
    "subject": "chemistry",
    "question_index": 1,
    "question": "Regarding compound 2-phenylpropene \\",
    "option": [
        "A. All atoms in the molecule lie in the same plane\\",
        "B. The main product of its hydration reaction is a secondary
        alcohol\\",
        "C. Complete combustion of 1 mol of this compound consumes 9
        mol O2\\",
        "D. It can undergo addition polymerization to form a polymer"
    ],
    "answer": "B",
    "images": {
        "question": ["2019-National-01"], "option": []
    }
}
```

3) Our dataset link on huggingface: Finally, we have constructed four datasets as follows: AP-exam-questions, TMUA-exam-questions, Alevel-exam-questions, and GaoKao-exam-question.

V. Results

This section appears to tend to suggest what might be characterized as the initial empirical findings of our study. What seems to be indicated, initially, is what appears to represent an initial comparative analysis of the macroscopic performance of several large language models across the

MMLU [4], LogiQA [6], and C-Eval [5] benchmarks. What also appears significant in this context, subsequently, is what appears to allow for delving into what seems to constitute the models' error patterns, through a quantitative error profile analysis, which tends to point toward what appears to be the specific strengths and weaknesses of each model. What this pattern seems to suggest, therefore, is that this analysis appears to provide a substantially empirical foundation for our eventual proposal of a hybrid model.

A. Comparative Performance on Standardized Benchmarks

Clearly, within this broader analytical model, what our limited evaluation appears to be revealing is what might be labeled as unique performance features for the leading large language models on these public benchmarks. What this, presumably, is reflecting, from analyzing the accuracy results shown in **Table II of the Appendix**, is that **Gemini 2.5 pro** and **GPT-5** seemed to be consistently hinting at what seems to be top accuracy across a majority of test subsets.

TABLE II. ACCURACY (%) OF LARGE LANGUAGE MODELS ON DIFFERENT STANDARDIZED TEST DATASETS.

Dataset	Sub-Dataset	GPT-5	Gemini 2.5 pro	ChatGLM-4.5	DeepSeek-R1
MMLU	college mathematics	98.00	99.00	85.00	84
	college physics	99.00	99.00	99.00	94.00
	college computer science	98.00	99.00	98.00	92.00
	high school geography	94.00	94.00	93.00	92.00
	high school mathematics	100.00	100.00	85.00	98.00
LogiQA	LogiQA	83.00	83.60	81.00	72.60
C-Eval	advanced mathematics	99.00	99.00	97.00	98.00
	college physics	91.00	90.00	93.00	90.00
	computer architecture	88.00	86.00	88.00	90.00
	computer network	94.00	98.00	90.00	100.00
	high school geography	89.00	92.00	86.00	91.00
	high school mathematics	94.00	97.00	96.00	87.00

What is also presumably noteworthy within these results, **ChatGLM-4.5** and **DeepSeek-R1** seemed to be showing what seems to be substantially comparable, and sometimes likely leading accuracy on certain specific professional knowledge subsets, like 'computer network' (DeepSeek-R1) and 'college physics' (ChatGLM-4.5). What these complicated theoretical relationships are presumably suggesting, what this result tends to be indicating, then, is that model accuracy is not necessarily monolithic. What, here, then, this analysis appears to be showing, is that each model typically seem to be exhibiting what seem to be "specializing" in different areas of knowledge. Given these methodological notes, what the error profiles from this benchmark analysis, shown in Fig. 3, further seem to be reflecting is what seems to be leaving discernible cognitive footprints, from DeepSeek-R1's what seems to be centered around a parsing-related error profile to ChatGLM-4.5 seeming to struggle with instruction-following, what this analyst finds especially interesting.

B. Quantitative Analysis of Error Profiles

To move beyond what might be characterized as macroscopic accuracy rates and to more deeply understand the root causes of model failures, we undertook what appears to be a detailed classification and quantitative analysis of all incorrect answers.

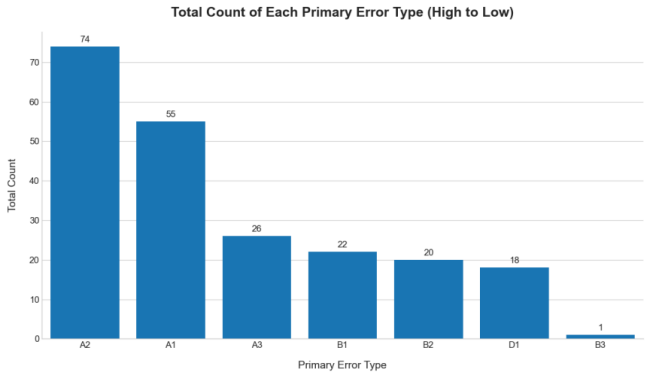
1) Error classification framework: Within this broader analytical framework, we tended to suggest what appears to be a systematic error classification framework, detailed ostensibly in **Table III of the Appendix**.

TABLE III. ERROR CLASSIFICATION SCHEME.

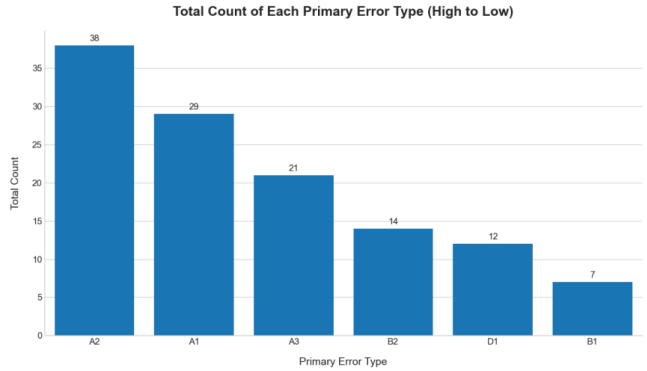
Code	Description
A.	Reading & Parsing
A1	Missing key words (e.g., NOT, EXCEPT)
A2	Misinterpreting entities/coreference
A3	CN-EN misunderstanding
B.	Computation & Formulas
B1	Arithmetic mistakes
B2	Algebra/formula errors
B3	Probability/statistics errors
C.	Reasoning
C1	Broken inference chain
C2	Confusing converse/inverse
C3	Overgeneralization/mismatch
D.	Strategy & Instruction-Following
D1	Output format violations
D2	Ignoring required method
D3	Hasty/insufficient verification
E.	Factuality & Hallucination
E1	Fabricated facts
E2	Misremembered knowledge
F.	Dataset/Label Issues
F1	Ambiguous problems
F2	Wrong official key

This framework appears to tend to suggest what seems to be six main categories (A–F), effectively covering multiple dimensions from text comprehension to instruction following. Given the complexity of these theoretical relationships, the priority order for error classification was D1 → A → B → C → E → F, what seems to generally indicate that we aimed to identify what appears to be the primary cause of failure.

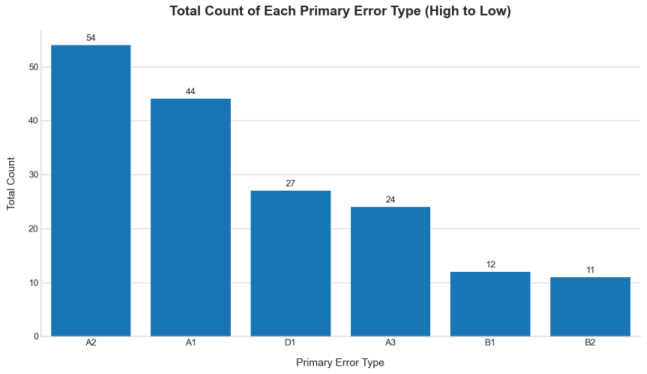
2) Model-specific error signatures: Based on what appears to be the framework outlined in Table III, we proceeded to label each model's incorrect answers and calculated what seems to be the distribution of primary error categories. The quantitative results are presented in a summary table in the Appendix, and the distributions are visualized in Fig. 3.



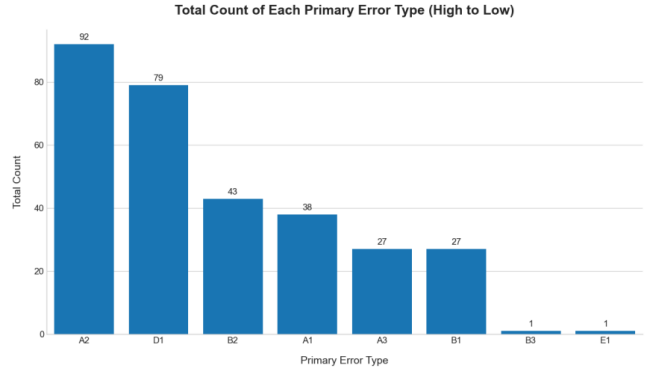
(a) DeepSeek error distribution



(b) Gemini error distribution



(c) GLM error distribution



(d) GPT error distribution

Fig. 3. Primary error distribution charts for the DeepSeek, Gemini, GLM, and GPT models. The charts illustrate the frequency of different error types as defined in Table 3.

What the analysis tends to support here appears to suggest each model's unique "cognitive fingerprint." What seems to emerge as a particularly significant finding, therefore, is that **Reading & Parsing (Category A)** appears to represent the predominant source of error for the majority of models, typically accounting for what seems to be the largest share of incorrect answers. What this tends to point toward what appears to be is that deep semantic understanding seems to generally constitute what appears to remain a substantially prevalent bottleneck for current large language models, what seems especially noteworthy in this analytical context.

However, within this broader framework, what we might find to be most interesting about them perhaps is that, beyond this general similarity, the models seem to tend to suggest what might be characterized as substantial and interesting differences on their secondary error patterns, which we attempt to describe below.

GPT-5. What the error profile of GPT-5 seems to generally reveal is what seems to be the most balanced distribution among the models. After what seems to be its primary weakness, what might be characterized as Reading & Parsing (A), its errors tend to be spread in what seems to be a relatively even distribution across **Reasoning (Category C)** and **Computation & Formulas (Category B)**. What this seems to suggest, given the complexity of these theoretical relationships, is that GPT-5 seems to be what might be characterized as a generalist model whose failures are not primarily concentrated in one particularly cognitive function, but seem to be spread across different types of complex tasks. What its errors in Category C, which includes broken inference chains (C1), seem to point towards is what seems to be an issue with long-range logical coherence, which certainly warrants further interpretive attention.

Gemini 2.5 pro. From this particular interpretive perspective, Gemini's error profile seems to be notably different than the other models, what seems to be most strikingly characterized by what seems to be a relatively noticeably higher proportion of Computation & Formulas (Category B) compared to the other models. While it also tends to struggle with Category A, what seems to be Gemini's tendency towards arithmetic (B1) and algebraic (B2) mistakes seems to point towards what may be a particularly interesting weakness in its numerical processing and symbolic manipulation capabilities, given its substantially robust performance on math benchmarks overall. What this tends to suggest, given the nuanced nature of these relationships, is that its high accuracy on math tends to seem to stem from what seems to be robust reasoning and problem setup rather than what might be considered perfectly accurate computation perhaps particularly noteworthy.

ChatGLM-4.5. What tends to come out as theoretically interesting, given these methodological considerations, is that, perhaps, the most remarkable aspect of ChatGLM-4.5's error profile seems to be what seems to be nothing other than a rather large rate of **Strategy & Instruction-Following (Category D)** error. It seemingly manages to fail more than what seems to be any other model on tasks seeming to require what seems to be strict adherence to output formats (D1) or methodology (D2). What this tends to point to is what seems to be a weakness with regards to constraint satisfaction and methodology adherence, even if its core reasoning (C) and computational (B) rates seem to be what seem to be comparatively low, what seems to render the standard interpretation confusing.

DeepSeek-R1. DeepSeek-R1 seems to tend to come across as suggesting what seems to be a highly specialised

error profile. What seems especially interesting in this context is that its failures tend to be predominantly concentrated in the realm of **Reading & Parsing (Category A)**. In stark contrast to the other models, its error rates for Computation (B), Reasoning (C) and InstructionFollowing (D) seem to be remarkably low. This distinctive signature, therefore, seems to point to what seems to be a model with what seems to be substantially reliable logical, mathematical and procedural execution capabilities, given how interesting the theoretical relationship in question seems to be, and its unique error pattern seems to point to what seems to be a model with what seems to be a significant weakness when it comes to initial comprehension of the prompt; if a problem seems to be largely correctly understood, it seems to be very likely to solve it largely correctly.

3) Performance simulation on full standardized examinations: What we then sought to explore, within this broader exploration, was to simulate a more realistic evaluation scenario in order to move beyond a benchmark of isolated questions, and have the two best performers, **GPT-5** and **Gemini 2.5 pro**, sit full examination papers from our dataset as though they were student candidates. What this experiment tends to assess, arguably, goes beyond knowledge, and seems to include what might be characterised as performance consistency and error patterns under conditions that tend to resemble what seem to be real-world academic tests, given the nature of the findings.

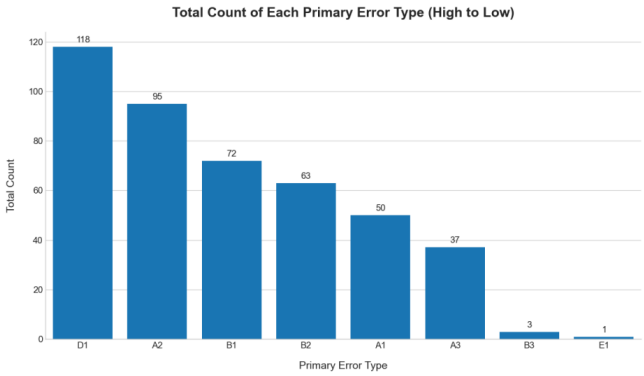
The accuracy results of our simulation can be seen in **Table IV of the Appendix**.

TABLE IV. ACCURACY (%) OF GPT-5 AND GEMINI 2.5 PRO ON FULL STANDARDIZED EXAMINATIONS.

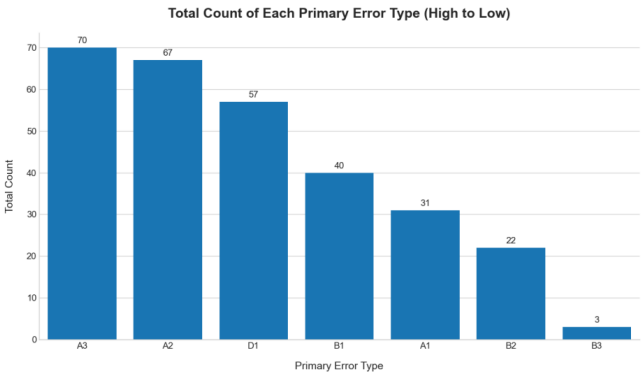
Examination	GPT-5 Accuracy (%)	Gemini 2.5 pro Accuracy (%)
Gaokao	88.50	93.37
AP	57.43	58.54
A-Level	81.99	96.46

What seems to emerge from these findings is that, seemingly unsurprisingly, both models seem to have achieved what seems to be impressive scores, and seem to frequently outperform the typical performer as would be expected of a human. What appears most interesting about these findings is that, DeepSeek-R1 seems to tend to come across as suggesting what seems to be a slight edge in the context of the Gaokao Math and TMUA exams, which seem to be known for what seems to be their complex reasoning and problem-solving depth, and Gemini seems to come across as tending to suggest a marginally better performer on the AP Calculus and A-Level Math exams, which seem to tend to place a strong emphasis on what tends to be curriculum-specific knowledge and procedural fluency, given the nature of this evidence.

What is perhaps even more interesting than the scores obtained, might be what appears to come out visually from the errors profiles obtained in this specific simulation scenario (what appears to be Figure 4 of the Appendix): when presented with a full examination, the latent strengths and weaknesses of the models tend to become much more apparent.



(a) GPT-5 error distribution (full exams)



(b) Gemini 2.5 pro error distribution (full exams)

Fig. 4. Error distribution charts for GPT-5 and Gemini 2.5 pro on the full standardized examination simulations. The charts illustrate the frequency of different error types as defined in Table 3.

- **GPT-5** was, under these particular regimes of exam conditions, suggestive of tending to exhibit a large clustering of errors of **Reasoning (Category C)** type, specifically 'Broken inference chain' (C1) more than would seem especially surprising, contextually. Which tends to indicate to me, at least, that while it may be good at individual problems it may seem to struggle more generally under the regime of having to maintain long-range logical consistency across what might be reasonably described as a multi-stage easy to extreme paper overall.
- **Gemini 2.5 pro** conversely seemed to be exhibiting a similar pattern to its rate of **Computation & Formulas (Category B)** type errors more than would seem especially surprising, contextually. Specifically its errors tended to more often be associated with 'Algebra/formula errors' (B2). Which tends to indicate to me, at least, given how messy the relationships are theoretically and how easy to get wrong in a multi-step high stakes problem like this, that the model is apparently more likely to get a calculation wrong, if given a correctly set up problem and reasoning.

What this head-to-head comparison, conducted from this particular interpretive perspective, within a realistic setting, appears to suggest is that even elite models appear to possess what might be characterized as distinct failure modes. GPT-5's weakness appears to be in sustained reasoning, while Gemini's appears to be in computational precision.

4) Synthesizing strengths and weaknesses: The case for a hybrid model: What the combination of the analysis of broad benchmarks (Appendix, Table 2 and Figure 3) with the targeted examination simulations (Appendix, Table 3 and

Figure 4) tends to point toward what appears to be is that it ostensibly appears that no single model is entirely omnipotent. Each model appears to tend to exhibit predominantly unique strengths and what appear to be critical weaknesses that are exposed given the multifaceted nature of this evidence, under a variety of different evaluation conditions. What this apparent complementarity seems to lend support to what may represent is substantially empirical evidence for constructing a more powerful and robust Hybrid Model. For example, what appears to be the reliable computational execution of a model like **DeepSeek-R1** could presumably be combined with what seems to be the advanced reasoning of **GPT-5**, while leveraging what tends to be the procedural strength of **Gemini 2.5 pro** to validate solutions. These findings appear to lay the groundwork for the hybrid model framework that we will explore in the next section.

VI. Discussion

What this study appears to tend to suggest is an ostensibly systematic assessment across multiple state-of-the-art large language models on standardized exams, what appears to show both promise and limitation predominantly in specific directions in terms of what these models seem to be capable of on what seem to be complex academic exams. What seems to be suggested by this work, extracted from analysis of performance on these exams, what seem to be broad benchmarks, in many cases simulations of what might be realistic full exams, is what seems to point towards what appear to be important results to extract with respect to LLMs in general. Furthermore, within this higher-level analysis, what seems to be suggested empirically is what appears to point towards what seems to be an important direction for future development. What these relationships seem to suggest is that this would appear to point towards an important focus on hybrid model architectures, given the complexity of these relationships.

A. The Landscape of LLM Performance

What these results seem to largely simulate, though, is a picture of what appears to be a relatively diverse special-purpose performance landscape. On general benchmarks, **Gemini 2.5 pro** and **GPT-5** seem to largely simulate, though, what appear to be top-ranked generalists, typically exhibiting relatively strong accuracy on a large number of subjects, what their strong performance likely indicates is a decent balance of knowledge recall and general reasoning ability.

What also seems important to note here, though, is that this general view appears to be considering the nuances of these results, what appears predominantly specialized models. What **DeepSeek-R1** seemingly achieving a perfect score in ‘computer network’ and **ChatGLM-4.5** seeming to exhibit relatively strong accuracy in ‘college physics’ seem to provide evidence that may support is that focused training can seemingly exhibit superior performance in specific domains, what seems to provide support to what may simulate a challenge to what appears to be a strong claim of being predominantly universally-generalist, what tends to emerge as a theoretical argument, Gemini’s superior accuracy on the A-Level and Gaokao exams, while GPT-5 leading on the reasoning-heavy TMUA, what these results seem to indicate is that performance is likely taskand structure-dependent.

What the seemingly large drop in accuracy for both models on the AP exams seems to indicate is that what seems to suggest is that both models seemingly share a vulnerability to certain styles of questions or specific curriculum nuances that aren’t well represented in their training data. What seems to require further interpretive consideration, within these parameters, seems to warrant further interpretive consideration.

B. Error Profiles as Cognitive Fingerprints

The most interesting takeaways of this study, what seems most noteworthy about these results, seem to come from the detailed error analysis, which seems to show what seems to be the clear “cognitive fingerprint” of each model. What seems to be most noteworthy coming out of these results, given the complexity of these theoretical relationships, is that the widespread presence of **Reading & Parsing (Category A)** errors seems to support what seems to be the notion that deep semantic understanding, deep grasp of the nuances and restrictions of a question, seems to endure as what seems to be a fundamental bottleneck for all current architectures. What seems most noteworthy coming out of these results, given the complexity of these theoretical relationships, is that models seem still prone to missing key negations or misinterpreting complex entity relationships beyond this shared weakness, and what the evidence seems to reveal, from this particular interpretive perspective, is that the models’ error profiles seem to differ markedly seemingly showing what seems to be the clear underlying architectural bias.

What seems to come out, under this broader analytical consideration, is that **DeepSeek-R1 and ChatGLM-4.5** seem to come out as what might be regarded as the specialists, since, respectively, DeepSeek’s absurdly low errors in computation and reasoning, and high rate of parsing errors, tend to push there towards what seems to be a clear case of a killer but poorly guided engine.

- What seems to come out, under these considerations, is that ChatGLM4.5’s high rate of **Instruction-Following (Category D)** errors seems to generally resolve to what seems to be a clear case of a model that seems to generally seem to have the ability to perform the core abilities, but generally seems to fail at following the constraints, what seems to be theoretically important here, seemingly resolving to a classic type of flaw that one would expect from standardized testing, seemingly a failure to generalize.
- What seems to come out from the data, under these evolving considerations, is that **GPT-5 and Gemini 2.5 pro** seem to come out as the elite generalists, and what seems to come out as a trade-off between the higher order cognitive skills. GPT-5’s clear weakness in **Reasoning (Category C)**, particularly in the full-exam sense, tends to push what seems to be a kind of “reasoning fatigue”, or rather, the idea that maintaining long, complex logical chains under stress seems to be the clear weak spot.
- What seems to come out from these considerations, regarding Gemini’s continued weakness in **Computation & Formulas (Category B)**, is that this seems to come out as a different kind of weakness, or rather, a clear preponderance against a sufficient level of precision in symbolic and numerical manipulation, especially when the overall reasoning is apparently solid.

What we think is most interesting about these different error signatures is that they don't just seem diagnostic; they seem more prescriptive in the sense that they can be described as something like "should be". What this pattern appears to be telling me is that getting GPT-5 to work seems to be less a matter of better calculation and more a matter of growing what seem to be stronger logical coherence, while getting Gemini to work seems generally to be a matter of needing more reliable computational abilities.

C. Limitations and Methodological Implications

However, in this broader analytical framework, the systematic error class classification framework we use here seems to provide what seems to be a useful way to move past simple accuracy scores. Because it seems to provide what seems to be a granular breakdown of failure modes, we can seem to more easily diagnose model weaknesses and seem to have what seems to be clear directions for improvement. However, for some limitations, we must ostensibly acknowledge given the complexity of these theoretical relationships. Question, perhaps, making traditional interpretations difficult is the subjectivity inherent in our manually labeled errors, while granular may perhaps introduce a subject bias. One of our model's limitations is that we evaluated students on multiple-choice questions to allow for objective scoring, but these models seemingly were not tested on subjective free-response questions. What seems to be important in this respect is that we seem not to have any graphical reasoning in our dataset, and thus, we ostensibly tested only text-based reasoning, developing what may be reliable, automatically graded rubrics for subjective questions and evaluating models on multimodal reasoning that seems to be a more holistic measure seems to be an important next step.

D. The Inevitable Path Towards Hybrid Models

And if anything, the overall result of this study tends to come to what seems like a pretty strong hint for a hybrid team that builds next-generation AI. What the numbers seem pretty clear on is that overall, there isn't a single model that's best. What we seem to have is a whole field of "homo sapiens cognitive engines" with different strengths and what seem to be predictable weaknesses. What that seems to point to is that the path to more robust intelligent systems doesn't seem to lie in finding the single "winner", but rather in architecture. How neat is the complementarity of the models? One can easily imagine a system where DeepSeek's reliable actor is simply told to execute by GPT-5 to parse a problem and solve it, possibly with Gemini's procedural power used as a final validation step. A system that would dynamically decide tasks based on the nature of the problem and known strengths of each multi-component part of the system, and in aggregate ostensibly be greater than the sum of its parts. This seems to be an empirical blueprint for designing cooperative intelligent systems that we seem to see in nature, taking the field from competing monolithic models to cooperative multi-part systems.

VII. Conclusion and Future Work

A. Conclusion

In this study, we systematically studied several state-of-the-art large language models in two contexts: standard benchmarks and simulated real-world standardized exams, and reached several series of interesting conclusions on the capabilities of state-of-the-art LLMs. Our study shows that there is no "omnipotent" model, but we can see a bunch of specialized "cognitive engines". First, elite models like **GPT-5** and **Gemini 2.5 pro** show strong capabilities across a vast range of knowledge with increasingly evident "cognitive fingerprints" (i.e., respectively, the weaknesses of maintaining complex long-chain inferences called "reasoning fatigue") under the continuous pressure of simulated exams, while that of Gemini demonstrates the weakness on precision of computational results and the stability of symbolic operations. Second, specialized models like **DeepSeek-R1** and **ChatGLM-4.5** also show more complementary capabilities. Specifically, DeepSeek shows very reliable logical and computational execution, but its main bottleneck is the front-end semantic understanding, while ChatGLM-4.5 shows a main weakness in strict instruction following, but this is a major weakness of standardized tests like ChatGLM, requiring a strict output format. Most importantly, we found that the most fundamental capability these models are most deficient in is deep semantic understanding. The appearance of "Reading & Parsing (Category A)" errors, although LLMs have achieved great successes in language generation, it is still very hard to accurately and flawlessly understand every constraint and nuance of a complex problem. In summary, the most important conclusion of this work is that the way to more powerful and robust AI is not to find the best model, but to be at the architectural level of synthesis and collaboration. The data and analysis we released have provided a solid empirical basis for the next generation of hybrid AI.

B. Future Work

Based on the findings and limitations of this study, we propose the following directions for future research:

- **Architectural Design and Implementation** Our main future work is to design and implement a dynamic hybrid model architecture. Specifically, we will try to design an intelligent assignment strategy to assign different stages of a certain task to different model components based on the type of problem (e.g., computation-intensive or reasoning-intensive) and the strengths of the models (e.g., reliable computational ability of DeepSeek or powerful reasoning ability of GPT-5). Designing effective collaboration methods among models will also be an interesting future work.
- **Improving Deep Semantic Understanding** Since category A errors are often produced, improving the deep semantic understanding of LLMs should be future research. This can be achieved by designing new model architectures, new pre-training tasks targeting at logical constraints and subtle semantics, or new fine-tuning methods to enhance the capability of parsing instructions.
- **Extending Dataset and Evaluation Metrics** Our aim is to have a more comprehensive evaluation of LLMs. It is challenging to overcome the limitations of our study. We will extend our dataset to include subjective questions (e.g., short-answer questions and essay questions) and further, we will conduct related research to design scoring rubrics that can objectively, automatically and reliably evaluate open-ended answers. In addition, when designing test sets, we will include more multimodal elements (e.g., chart images and formula images) to target cross-modal reasoning.
- **Conducting Longitudinal Study** Since large models will keep iterating, it is interesting and valuable to

conduct a longitudinal tracking study. We will periodically re-evaluate new versions of mainstream models using our standardized examination dataset to track the evolution of their "cognitive fingerprints," and further, we will analyze whether their capabilities are enhanced in a balanced way, whether their old weaknesses are solved and whether new weaknesses have emerged.

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